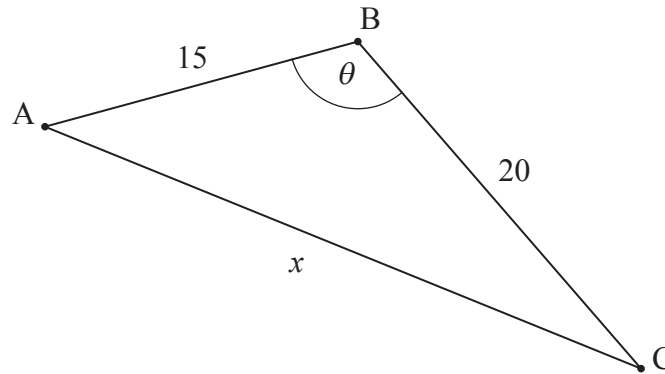


9. [Maximum mark: 7]

Consider triangle ABC with $AB = 15\text{ cm}$, $BC = 20\text{ cm}$, $AC = x\text{ cm}$ and $\hat{ABC} = \theta$ where $0 < \theta < \frac{\pi}{2}$.

diagram not to scale



Angle θ is decreasing at the constant rate of $\frac{\pi}{60}$ radians per minute.

Determine how fast the length of AC is changing when the area of the triangle is 140 cm^2 .

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